MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, AND GATES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, Advanced Schottky TTL, positive AND logic gate microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quadruple, 2-input positive AND gate
02	Triple, 3-input positive AND gate

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	C-2A (20-terminal, .350" x .350"), square chip carrier package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.2 V at -18 mA to +7.0 V
Storage temperature range	-65°C to +150°C
Maximum power dissipation per device(P_D) 1/	
Device type 01	71 mW
Device type 02	53 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C, D, X, and 2	(See MIL-M-38510, appendix C)
Junction temperature (T_J) 2/	175°C

1/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
2/ Maximum junction temperature shall not be exceeded except in accordance with allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

1.4 Recommended operating conditions.

Supply voltage	4.5 minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.8 V
Normalized fanout (each output) 3/	
Low logic level	33 maximum
High logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3/ Device will fanout in both high and low levels to the specified number of data inputs on the same device type as that being tested.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -1.0 mA, V_{IH} = 2.0 V	All	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 20 mA, V_{IL} = 0.8 V, V_{IH} = 2.0 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.2	V
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 7.0 V	All		100	µA
Low-level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	All	-.03	-.60	µA
Short-circuit output current 1/	I_{OS}	V_{CC} = 5.5 V V_O = 0.0 V	All	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{IN} = 0.0 V, V_{OUT} = 2.5 V	All	60		mA
High-level supply current	I_{CCH}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01		8.3	mA
			02		6.2	mA
Low-level supply current	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 0.0 V	01		12.9	mA
			02		9.7	mA
Propagation delay time, high-to-low level	t_{PHL}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500Ω ±5%	All	2.0	7.5	ns
Propagation delay time, low-to-high level	t_{PLH}	V_{CC} = 5.0 V, C_L = 50 pF ±10%, R_L = 500Ω ±5%	All	2.5	7.5	ns

1/ Not more than one output should be shorted at a time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 8 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,9, 10,11	1*,2,3,9
Group A test requirements (method 5005)	1,2,3, 9,10,11	1,2,3,9 10,11
Group B test requirements (method 5005), (subgroup 5)	1,2,3, 9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspections. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3 or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:19Z
Check the source to verify that this is the current version before use.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 7, and 8 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

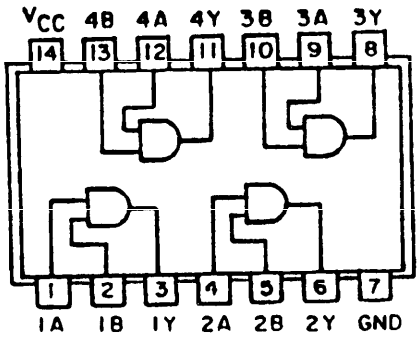
6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for Government microcircuit applications (original equipment) and logistic purposes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

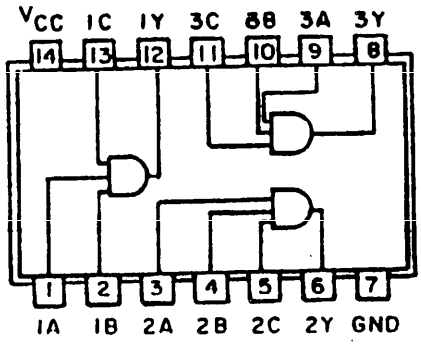
Device type 01

Cases A, B, C, and D



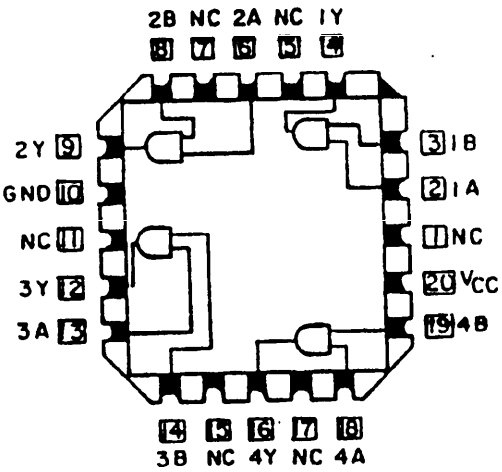
Device type 02

Cases A, B, C, and D



Device type 01

Cases X and 2



Device type 02

Cases X and 2

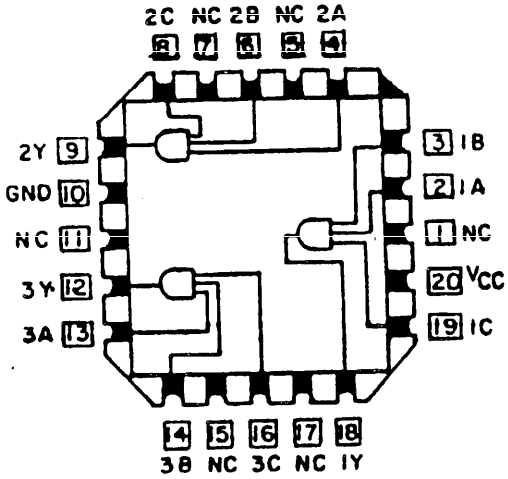


FIGURE 1. Terminal connections and logic diagrams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

Device type 01

Truth table (each gate)		
Inputs		Output
A	B	Y
L	L	L
H	L	L
L	H	L
H	H	H

Positive logic Y = AB

Device type 02

Truth table (each gate)			
Inputs			Output
A	B	C	Y
L	L	L	L
H	L	L	L
H	H	L	L
H	H	H	H

Positive logic Y = ABC

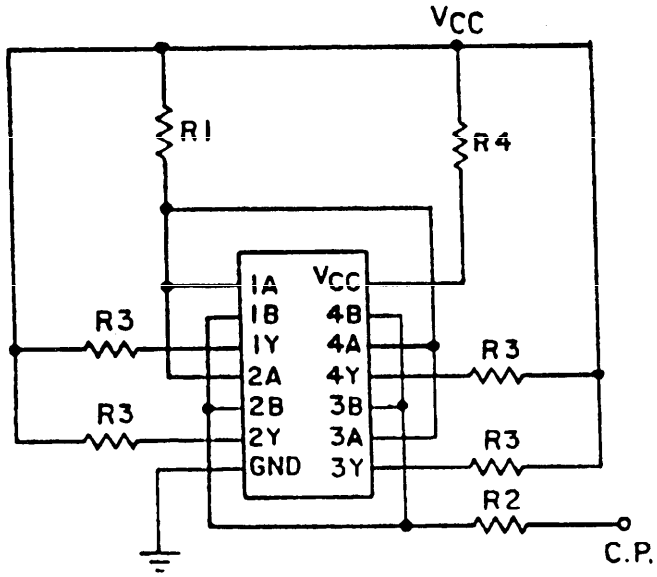
FIGURE 2. Truth tables and logic equations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

Device type 01

Cases X and 2

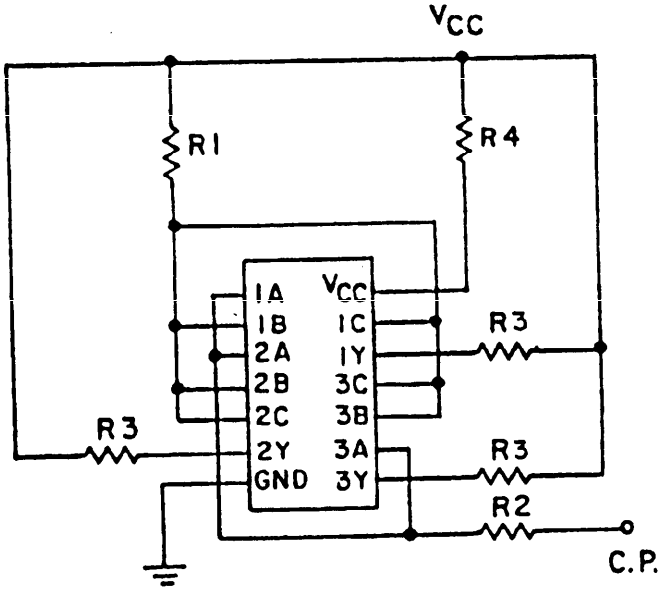
(Pins not referenced for cases X and 2 are N/C)



Device type 02

Cases X and 2

(Pins not referenced for cases X and 2 are N/C)

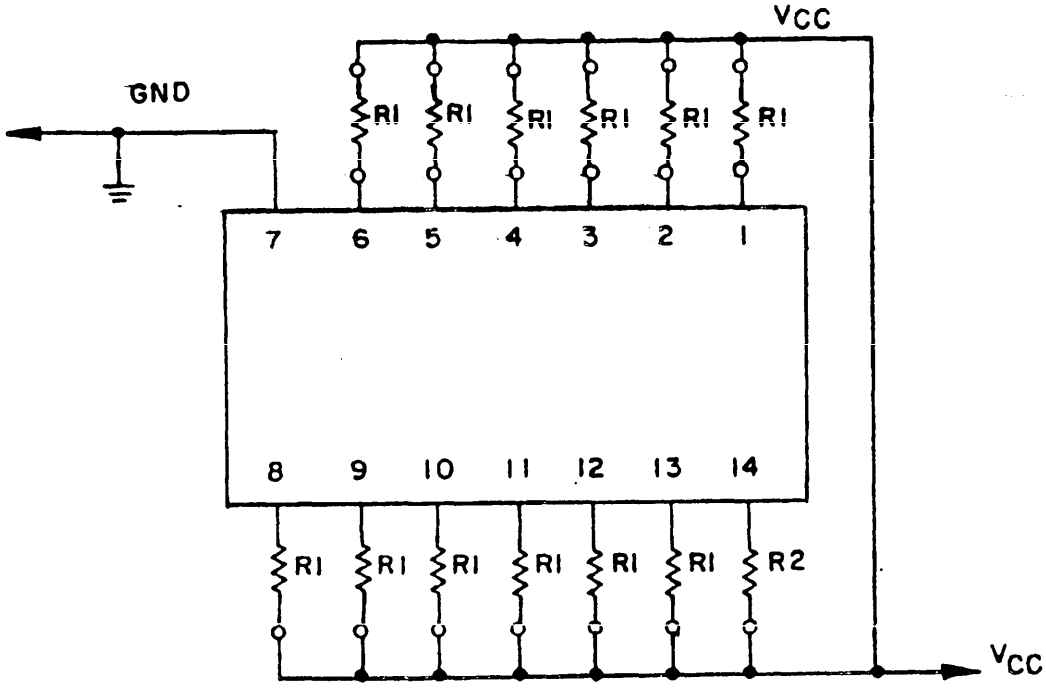


NOTES:

1. R1 = 1 kΩ ±5 percent.
   R2 = 160Ω ±5 percent device 01; device 02 R2 = 220Ω ±5 percent.
   R3 = 270Ω ±5 percent .
2. V_{CC} and R4 shall be chosen to insure a minimum of 5.5 V at the input of the device V_{CC} terminal.
3. C.P. = 100 kHZ ±50 percent square wave; duty cycle = 50 ±15 percent, V_{IL} = -0.5 minimum to 0.8 V maximum; V_{IH} = 2.0 minimum to 5.5 V maximum.

FIGURE 3. Burn-in and life test circuits.

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:19Z
Check the source to verify that this is the current version before use.

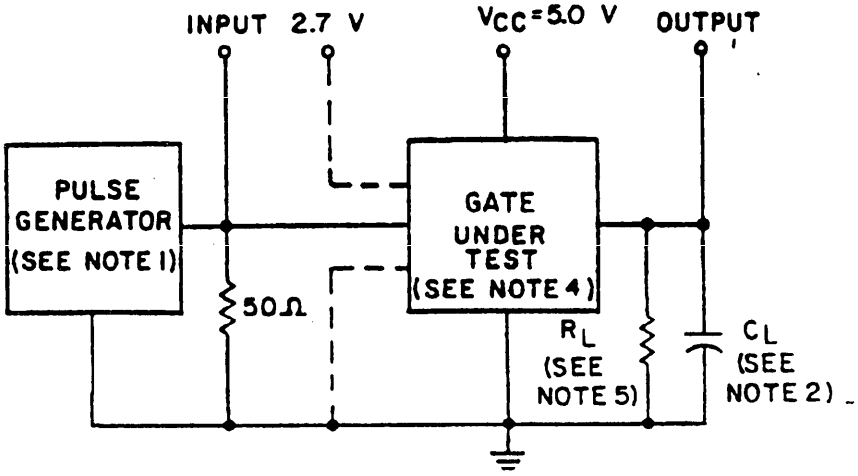


NOTES:
1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2kΩ, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

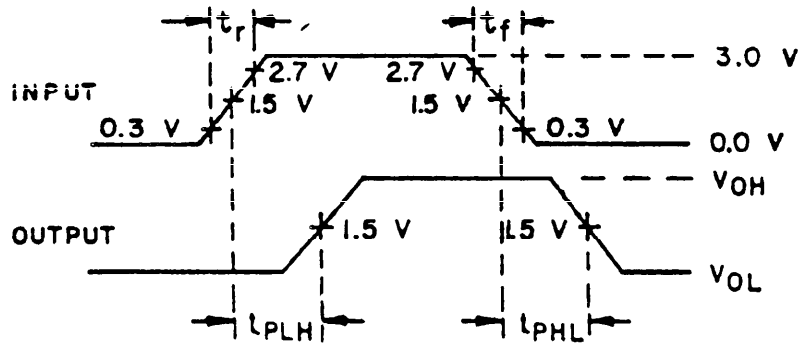
FIGURE 3. Burn-in and life test circuits -Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

TEST CIRCUIT



TEST CIRCUIT



NOTES:
1. Pulse generator has following characteristics:
   $t_r = t_f \leq 2.5$ ns; PRR $\leq$ 1 MHz, and $Z_{OUT} \cong 50\Omega$.
2. C_L = 50 pF ±10 percent including scope, probe, wiring, and stray capitance, without package in test fixture.
3. Voltage measurements are to be made with respect to network ground terminal.
4. Inputs not under test are at 2.7 V or GND as specified in table III.
5. R_L = 500Ω ±5 percent.

FIGURE 4. Switching time test circuit and waveforms (all devices).

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:19Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J/ 3,2 Case A,B,C,D	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Limits Min	Limits Max	Unit
					1A	1	1B	2	2A	2	GND	3A	2A	3	4Y	4	3A	3	4B	4	4Y	4A	4B	V_{CC}				
1 T_C = 25°C	V_{OL}	3007		1	0.8 V 2.0 V		2.0 V 0.8 V		0.8 V 2.0 V		GND		20 mA 20 mA		20 mA 20 mA		0.8 V 2.0 V		2.0 V 0.8 V		20 mA 20 mA		0.8 V 2.0 V	4.5 V	17 27 37 47		0.5	V
	V_{OH}	3006		9	2.0 V		2.0 V		2.0 V		GND		-1 mA		-1 mA		2.0 V		2.0 V		-1 mA		2.0 V		17 27 37 47	2.5		V
	V_{IC}			13	-10 mA		-10 mA		-10 mA		GND		-10 mA				-10 mA		-10 mA				-10 mA	5.5 V	15 25 35 45 48		-1.5	V
	I_{IH1}	3010		21	2.7 V	GND	2.7 V	GND	2.7 V	GND	GND		2.7 V	GND			2.7 V	GND	2.7 V	GND		2.7 V	GND	5.5 V	15 25 35 45 48		20	μA
	I_{IH2}	3010		29	7.0 V	GND	7.0 V	GND	7.0 V	GND	GND		7.0 V	GND			7.0 V	GND	7.0 V	GND		7.0 V	GND		15 25 35 45 48		100	
	I_{IL1}	3009		37	0.5 V 5.5 V		0.5 V 5.5 V		0.5 V 5.5 V		GND						0.5 V 5.5 V		5.5 V 0.5 V			0.5 V 5.5 V			15 25 35 45 48		2/	μA
	I_{OS}	3011		45	5.5 V		5.5 V		5.5 V		GND		0.0 V		0.0 V		5.5 V		5.5 V		0.0 V	5.5 V			17 27 37 47	-60	-150	mA
	I_{DD}			49 50 51 52	GND				GND		GND		2.5 V		2.5 V		GND				2.5 V	GND		4.5 V	17 27 37 47	60		mA
	I_{CCH}	3005		53	5.5 V		5.5 V		5.5 V		GND						5.5 V		5.5 V			5.5 V		5.5 V	V_{CC}		6.3	mA
	I_{CCL}	3005		54	GND		GND		GND		GND						GND		GND			GND		5.5 V	V_{CC}		12.9	mA

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

'active low' chip Enable inputs. NOTE: Address and enable inputs must b

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ K,Z	2	3	4	6	8	9	10	12	13	14	15	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14			Min	Max	
			Test no.	1A	1B	1Y	2A	2B	2Y	GND	3Y	3A	3B	4Y	4A	4B	V_{CC}				
1	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
9 T_C = 25°C	t_{PHL}	3003 Fig. 4	55 56 57 58 59 60 61 62	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	GND	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	5.0 V	1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	2.5	5.3	ns
	t_{PLH}	3003 Fig. 4	63 64 65 66 67 68 69 70	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT		OUT OUT	IN 2.7 V	2.7 V IN	OUT OUT	IN 2.7 V	2.7 V IN		1A to 1Y 1B to 1Y 2A to 2Y 2B to 2Y 3A to 3Y 3B to 3Y 4A to 4Y 4B to 4Y	3.0	6.6	
10	t_{PHL}	3003 Fig. 4	71-78	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C																2.0	7.0	ns
	t_{PLH}		79-86																	2.5	7.0	
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

1/ Pins not referenced (i.e., 1, 5, 7, 11, 15, and 17) are N/C.

2/ I_{IL} limits shall be as follows:

Parameter	Limits in mA for circuit		
	A	B	C
I_{IL1}	-.25/-.80	-.037/-.80	-.037/-.80

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ X,Z	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}		Min	Max	
			Test no.	1A	1B	2A	2B	2C	2Y												
T_C = 25°C 1	V_{OL}	3007	1	0.8 V	2.0 V					GND					20 mA	2.0 V	4.5 V	1Y		0.5	V
			2	2.0 V	0.8 V											2.0 V		1Y			
			3	2.0 V	2.0 V											0.8 V		1Y			
			4			0.8 V	2.0 V	2.0 V	20 mA									2Y			
			5			2.0 V	0.8 V	2.0 V										2Y			
			6			2.0 V	2.0 V	0.8 V										2Y			
			7								20 mA							3Y			
			8									20 mA	0.8 V	2.0 V	2.0 V			3Y			
			9										2.0 V	0.8 V	2.0 V			3Y			
	V_{OH}	3006	10	2.0 V	2.0 V										-1 mA	2.0 V		1Y	2.5		V
			11			2.0 V	2.0 V	2.0 V	-1 mA									2Y			
			12									-1 mA	2.0 V	2.0 V	2.0 V			3Y			
	V_{IC}		13	-18 mA														1A		-1.2	V
			14		-18 mA													1B			
			15			-18 mA												2A			
			16				-18 mA											2B			
			17					-18 mA										2C			
			18										-18 mA					3A			
			19											-18 mA				3B			
			20												-18 mA			3C			
			21													-18 mA		1C			
	I_{IH3}	3010	22	2.7 V	GND											GND	5.5 V	1A		20	μA
			23	GND	2.7 V											GND		1B			
			24	GND	GND											2.7 V		2A			
			25			2.7 V	GND	GND										2B			
			26			GND	2.7 V	GND										2C			
			27			GND	GND	2.7 V										2C			
			28										2.7 V	GND	GND			3A			
			29										GND	2.7 V	GND			3B			
			30										GND	GND	2.7 V			3C			
	I_{IH2}	3010	31	7.0 V	GND											GND	5.5 V	1A		100	μA
			32	GND	7.0 V											GND		1B			
			33	GND	GND											7.0 V		2A			
			34			7.0 V	GND	GND										2B			
			35			GND	7.0 V	GND										2C			
			36			GND	GND	7.0 V										2C			
			37										7.0 V	GND	GND			3A			
			38										GND	7.0 V	GND			3B			
			39										GND	GND	7.0 V			3C			
	I_{IL1}	3009	40	0.5 V	5.5 V											5.5 V	5.5 V	1A	2/	2/	μA
			41	5.5 V	0.5 V											5.5 V		1B			
			42	5.5 V	5.5 V											0.5 V		2A			
			43			0.5 V	5.5 V	5.5 V										2B			
			44			5.5 V	0.5 V	5.5 V										2C			
			45			5.5 V	5.5 V	0.5 V										2C			
			46										0.5 V	5.5 V	5.5 V			3A			
			47										5.5 V	0.5 V	5.5 V			3B			
			48										5.5 V	5.5 V	0.5 V			3C			
	I_{OS}	3011	49	5.5 V	5.5 V											0.0 V	5.5 V	1Y	-40	-150	mA
			50			5.5 V	5.5 V	5.5 V	0.0 V									2Y			
			51								0.0 V	5.5 V	5.5 V	5.5 V	5.5 V			3Y			
	I_{OD}		52	GND												2.5 V		1Y	40		μA
			53			GND			2.5 V								4.5 V	2Y			
			54								2.5 V	GND						3Y			
	I_{CCH}	3005	55	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	V_{CC}		6.2	mA
	I_{CCL}	3005	56	GND	GND	GND	GND	GND				GND	GND	GND		GND	5.5 V	V_{CC}		9.7	mA

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases 1/ E,Z	2	3	4	6	8	9	10	12	13	14	15	16	18	19	20	Measured terminal	Limits		Unit
			Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14			Min	Max	
			Test no.	1A	1B	2A	2B	2C	2Y	GND	3Y	3A	3B	3C	1Y	1C	V_{CC}					
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
9 T_C = 25°C	t_{PHL}	3003 Fig. 4	57 58 59 60 61 62 63 64 65	IH 2.7 V 2.7 V -	2.7 V IH 2.7 V	I_{IH} 2.7 V	2.7 V IH 2.7 V	2.7 V 2.7 V IH	OUT •	GND • • • • • • •	OUT •	IH 2.7 V 2.7 V	2.7 V IH 2.7 V	2.7 V 2.7 V IH	OUT •	2.7 V 2.7 V IH	5.0 V • • • • • • •	1A to 1Y 1B to 1Y 1C to 1Y 2A to 2Y 2B to 2Y 2C to 2Y 3A to 3Y 3B to 3Y 3C to 3Y	2.5 • • • • • • • •	5.0 • • • • • • • •	ns • • • • • • • •
	t_{PLH}	3003 Fig. 4	66 67 68 69 70 71 72 73 74	IH 2.7 V 2.7 V	2.7 V IH 2.7 V	IH 2.7 V 2.7 V	2.7 V IH 2.7 V	2.7 V 2.7 V IH	OUT •	•	OUT •	IH 2.7 V 2.7 V	2.7 V IH 2.7 V	2.7 V 2.7 V IH	OUT •	2.7 V 2.7 V IH	• • • • • • • •	1A to 1Y 1B to 1Y 1C to 1Y 2A to 2Y 2B to 2Y 2C to 2Y 3A to 3Y 3B to 3Y 3C to 3Y	3.0 • • • • • • • •	5.0 • • • • • • • •	• • • • • • • • •
10	t_{PHL}	3003 Fig. 4	75-83	Same tests and terminal conditions as for subgroup 9, except T_C ≤125°C.																2.0	7.5	•
	t_{PLH}		84-92																	2.5	7.5	•
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																					

1/ Pins not referenced (i.e., 1, 5, 7, 11, 15, and 17) are N/C.

2/ I_{IL} limits shall be as follows:

Parameter	Limits in mA for circuits		
	A	B	C
I_{IL1}	-.25/-.60	-.03/-.60	-.03/-.60

Source: https://assist.dla.mil - Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Ground zero voltage potential

V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal

I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead and material finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-35810 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F08
02	54F11

6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Manufacturer		
	Circuit A	Circuit B	Circuit C
	Fairchild	Motorola	Signetics
01	X	X	
02	X	X	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:19Z
Check the source to verify that this is the current version before use.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Air Force - 11, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1103)

Source: https://assist.dla.mil -- Downloaded: 202610-21T14:19Z
Check the source to verify that this is the current version before use.